| INCH-POUND |

MIL-C-28628C(YD)
8 July 1991
SUPERSEDING
MIL-C-28628B(YD)
26 November 1984

MILITARY SPECIFICATION

CAMEL, WOOD, MARINE; SINGLE LOG CONFIGURATION, UNTREATED AND TREATED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single log configuration marine wood camels, treated and untreated, hereinafter referred to as log camels.

* 1.2 Classification. Log camels shall be of the following types and classes, as specified (see 6.2):

Type I - Treated.

Class 1 - Moderate borer hazard.
Class 2 - Severe borer hazard.
Class 3 - Dual treatment.

Type II - Untreated.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1945

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2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

DOD-P-23236 - Paint Coating Systems, Steel Ship Tank, Fuel and Salt Water Ballast.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Navy Publication & Printing, Service Office, Building 4D, NPM DODSSP, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

ASTM

ASTM A 36 - Structural Steel.
ASTM A 53 - Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
ASTM A 386 - Zinc Coating (Hot Dip) on Assembled Steel Products.
ASTM D 25 - Round Timber Piles.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Rd., Miami, FL 33135.)

AMERICAN WOOD-PRESERVERS' ASSOCIATION (AWPA)

C1 - All Timber Products - Preservative Treatment by Pressure Process.
C3 - Piles - Preservative Treatment by Pressure Process.

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(Application for copies should be addressed to the American Wood-Preservers' Association, P.O. Box 849, Stevensville, MD 21666.)

AMERICAN WOOD PRESERVERS BUREAU (AWPB)

AWPB MP-1 - Dual Treatment of Marine Piling Pressure Treated with Water-Borne Preservatives and Creosote for use in Marine Waters.
AWPB MP-2 - Marine Piling Pressure Treated with Creosote for use in Marine Waters.
AWPB MP-4 - Marine Piling Pressure Treated with Water-Borne Preservatives for Use in Marine Waters.

(Application for copies should be addressed to the American Wood Preservers Bureau, P.O. Box 6085, 2772 South Randolph Street, Arlington, VA 22206.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Log camels shall consist of natural wood logs over 20 inches in diameter at the butt. Type I log camels shall be treated as specified (see 3.6.1 and 6.2), and type II shall be untreated. When specified (see 6.2 and 6.6), the logs shall be lathe turned to a uniform nominal diameter along the entire length of the log.

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.2.1 Wood for log camels. Since difficulty may be encountered in penetration and retention, it is the responsibility of the contractor to select logs for treatment that have sufficient sapwood thickness to permit obtaining the penetration and retention required, or otherwise ensure the required penetration and retention are achieved (see 3.5). Logs shall be cut above ground swell from sound live trees and shall be peeled free from bark soon after cutting. The wood shall be sound and free from decay, red heart, or insect attack except insect damage consisting of holes 1/16 inch or less in diameter or

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surface scoring or channeling is permitted. Logs used for camels shall conform to ASTM D25, except as modified by this specification.

3.2.1.1 Species. Treated (type I) log camels shall be of either Coastal Douglas Fir or Southern Pine. Alternative species used for untreated (type II) log camels shall retain their shape, have adequate buoyancy (see 3.2.1.5), and have fiber strength and density similar to the treated, type I camel log species.

3.2.1.2 Scars. Logs may have sound turpentine scars which are undamaged by insects. No scars shall exceed 3 feet in length.

3.2.1.3 Foreign material. Logs shall be free of all nails, spikes, and other metal except as specified herein.

3.2.1.4 Straightness. The distance from the center of a cross section to a straight line from center of butt to center of tip shall not exceed 1 inch per 10 feet of log length.

3.2.1.5 Buoyancy. When tested as described in 4.4, the freeboard (that portion of the log above the water surface) of each log, prior to treatment, shall be no less than 20 percent of the log's diameter.

3.2.2 Pipe. Pipe for hawse pipes shall be 4-inch double extra strong pipe conforming to ASTM A53.

3.2.3 Steel plate. Steel plate for hawse pipe flange shall conform to ASTM A36.

3.3 Size. Log camels shall be specified by length and either diameter or circumference. Log camels of natural taper shall be specified by minimum butt circumference and lathe turned logs shall be specified by diameter.

3.3.1 Length. Log camels shall be 30, 35, 40, or 50 feet in length as specified (see 6.2), and have a tolerance of +1 foot, -0 foot. Special lengths, when specified (see 6.2), shall be specified in 2-foot multiples under 40 feet and 5-foot multiples over 40 feet. Ends shall be cut square to the centerline.

3.3.2 Cross section.

3.3.2.1 30, 40 and 50 foot logs. The nominal minimum butt circumference of 30, 40, and 50 foot log camels shall be 63, 75, or 87 inches as specified (see 6.2). All circumference measurements are taken under the bark. Tip circumference shall not be less than the values given in table I.

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TABLE I. Specified butt circumference with minimum tip circumference (diameter approximated).

Length (in/feet)	Required minimum butt circumference (inches) 63 (20 Dia)	75 (24 Dia)	87 (28 Dia)
	Minimum allowable tip circumference		
30	47 (15 Dia)	59 (19 Dia)	72 (23 Dia)
40	42 (13 Dia)	54 (17 Dia)	67 (21 Dia)
50	37 (11 Dia)	49 (15 Dia)	62 (19 Dia)

3.3.2.2 35 foot logs. Unless otherwise specified (see 6.2), 35 foot log camels shall have a minimum tip diameter of 35 inches. The butt diameter shall be not more than 10 percent greater than the tip diameter. The equivalent diameter, when turned on a lathe, shall be approximately 14 inches.

3.3.2.3 Lathe turned log camels. When lathe turned log camels are specified, the diameter of any perpendicular cross section on the log shall not vary by more than +/-1 inch from the specified nominal diameter.

3.3.3 Roundness. The ratio of the maximum to minimum diameter at any perpendicular cross section shall not exceed 1:2.

3.4 Bark removal.

3.4.1 Type I. Types I log camels shall be clean peeled. Clean peeled logs shall have at least 80 percent of the inner bark, well distributed over the entire surface of the log, removed, and shall have no strip of inner bark larger than 1 inch by 6 inches remaining.

3.4.2 Type II. Type II log camels shall be either rough peeled or unpeeled as specified (see 6.2). Rough peeled logs require the complete removal of all outer bark, while unpeeled logs require no bark removal.

3.5 Incising. Unless otherwise specified (see 6.2), incising of type I log camels shall be permitted before treatment (see 3.2.1). When lathe turned logs are specified, they shall be full length incised to accommodate the loss of sapwood and permit the required penetration and retention of preservative treatment (see 6.7).

3.6 Treatment.

3.6.1 Wood preservative treatment. Treatment for type I log camels shall be class 1, 2, or 3 as specified (see 6.2) and shall be performed after holes for hawse pipes have been drilled. Preservative treatments and minimum retention for each class shall conform to the applicable requirements of the AWPA

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standards for treatment and to the AWPB standard for quality control listed for the class in table II.

3.6.1.1 Certification. The AWPB quality mark for the class of log camel shall be applied to each treated log in accordance with the applicable AWPB standard (see table II) and shall be an indication of conformance to the requirements of treatment and retention for type I log camels.

TABLE II. Treatment for Type I Camel Logs.

Class	Hazard	Treatment	AWPA Standard for Treatment	AWPB Standard Quality Control
1	Moderate borer hazard (Teredo and Pholad present, no limnoria tripunctata)	Creosote or creosote coal tar	C1 and C3 (for marine piles)	MP-2
2	Severe borer hazard (Teredo and limnoria tripunctata present, no Pholads)	ACA*, CCA*, or ACZA*	C1 and C3 (for marine piles)	MP-4
3	Dual treatments (Limnoria tripunctata and Pholads present)	First treatment ACA, CCA, or ACZA. Second treatment creosote or creosote coal-tar.	C1 and C3 (for marine piles)	MP-1

*ACA, CCA, and ACZA are abbreviations for Amoniacal Copper Arsenate Chromated Copper Arsenate and Ammoniacal Copper Zinc Arsenate, respectively.

3.6.2 Treatment of steel. Steel used in construction of hawse pipes shall be hot dip galvanized in accordance with ASTM A386. Galvanized areas damaged by welding shall be cleaned, primed, and painted with a 12 mil dry film thickness coating of type 1, class 1, epoxy paint in accordance with DOD-P-23236.

3.7 Hawse pipes. When specified (see 6.2), hawse pipes shall be installed in each end of log camels. Unless otherwise specified (see 6.2 and 6.6), hawse pipes shall conform to the requirements listed herein and to figure 1. Pipe shall be force-fit into a hole drilled on the floating vertical centerline of the camel log. The floating vertical centerline shall be determined and marked during the buoyancy test of 4.4. The floating vertical centerline is defined as a line passing through the center of the log's cross sectional area and perpendicular to the water's surface when the log is at rest. While the log is at rest, the location of the waterline and the floating vertical centerline shall be noted. Seven and 1/2-inch diameter by 1/2-inch thick steel plate washers shall be welded as flanges to each end of the pipe, as shown in figure 1, after insertion of the pipe in the log. Welding shall be a continuous groove weld conforming to 3.8.1. The hawse pipes shall be located at a distance of 20 percent of the total log length from each end of the camel.

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3.8 Workmanship.

* 3.8.1 Welding. Welding procedures shall be in accordance with AWS D1.1. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.2.1).
b. Packaging inspection (see 4.5).

4.2.1 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and the buoyancy test of 4.4. Unless otherwise specified (see 6.2), the inspection will be performed at the point of production.

4.3 Examination. Each log camel shall be examined for compliance with the requirements in section 3 of this specification. This element of inspection shall encompass all visual examinations, dimensional measurements, review of certificates of inspection, and check for the AWPB quality mark. Any log camel having one or more defects shall be cause for rejection of that camel.

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4.3.1 Examination before treatment. Type I log camels shall be examined for compliance with the requirements of 3.2 through 3.4.2, within 10 days prior to treatment.

4.3.2 Examination after treatment. After preservative treatment, and installation of hawse pipes if required, type I log camels shall be examined for compliance with 3.5 through 3.8.1.

4.4 Buoyancy test. Prior to treatment, each log shall be placed in water and allowed to float freely. The log shall conform to the buoyancy requirements of 3.2.1.5.

4.5 Packaging inspection. The marking of log camels shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

* 5.1 Packing. The log camels shall be prepared for shipment in a manner which will insure that delivery shall comply with applicable carrier rules and regulations.

5.2 Marking. Log camels shall be marked in accordance with MIL-STD-129 in addition to the AWPB quality mark as specified in 3.6.1.1.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Log camels are intended for use as a floating device to breast out a vessel from a pier or mooring, or to separate two vessels to provide protection to the vessels and minimize damage to pier pilings. The log camels covered in this specification are for fresh-water or sea water use.

6.1.1 Type I, class 1. Type I, class 1 log camels are for use in marine waters in those areas of moderate borer hazard where limnoria tripunctata attack is limited or not expected (primarily in northern latitudes but with exceptions in both northern and southern latitudes).

6.1.2 Type I, class 2. Type I, class 2 log camels are for use in marine waters in those areas of severe borer hazard where limnoria tripunctata attack is known or expected and where pholad attack is absent.

6.1.3 Type I, class 3. Type I, class 3 log camels are for use in marine waters in those areas of severe borer hazard where limnoria tripunctata and pholad attack is known or expected.

6.1.4 Type II. Type II, untreated log camels are for use in fresh water, where there is no marine borer hazard.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of log camels required (see 1.2, 3.1, and 3.6.1).
c. When referenced documents shall be other than the issue listed in that issue of the DoDISS specified in the solicitation (see 2.1.1).
d. When logs shall be lathe turned to uniform nominal diameter (see 3.1).
e. Length(s) of log camel(s) (see 3.3.1).
f. Nominal butt circumference for 30, 40, or 50 foot log camels (see 3.3.2.1).
g. When diameter of 35 foot logs is other than specified (see 3.3.2.2).
h. Whether type II log camels shall be rough peeled or unpeeled (see 3.4.2).
i. When incising of type I camel log before treatment shall not be permitted (see 3.5).
j. Treatment for type 1 log camels (see 3.6.1).
k. When hawse pipes shall be installed (see 3.7).
l. When hawse pipes shall be other than as specified (see 3.7).
m. When quality conformance inspection shall be performed at other than the point of production (see 4.3).

* 6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.4 Related documents. Additional documents which are not directly referenced, but may be required for review or in the performance of a contract are listed below:

AWPA A1 - Analysis of Creosote and oil-Type Preservatives.
AWPA A2 - Analysis of Water Borne Preservatives and Fire Retardant Formulations.
AWPA A4 - Sampling Wood Preservatives.
AWPA A6 - Determination of Oil-Type Preservatives and Water in Wood.
AWPA A7 - Wet Ashing Procedure for Preparing Wood for Chemical Analysis.
AWPA A8 - Qualitative Recovery of Creosote or Creosote-Coal Tar Solution from Freshly Treated Piles, Poles or Timbers (Squeeze Method).
AWPA A9 - Analysis of Treated Wood and Treating Solutions by X-ray Emission Spectroscopy.
AWPA A10 - Analysis of CCA Treating Solutions and CCA Treated Wood by Colorimetry.
AWPA A11 - Analysis of Treated Wood and Treating Solutions by Atomic Absorption Spectroscopy.
AWPA M2 - Inspection of Treated Timber Products.
AWPA M6 - Brands Used on Forest Products.
AWPA P1 - Coal Tar Creosote for Land and Fresh Water Use.
AWPA P2 - Creosote-Coal Tar Solutions.

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AWPA P12 - Creosote-Coal Tar Solutions to he used in the Treatment of Marine (Coastal Waters) Piles and Timbers.
AWPA P13 - Coal Tar Creosote to be used in the Treatment of Marine (Coastal Waters) Piles and Timbers.

6.5 Subject terms (key word) listing.

Buoyancy
Coastal Douglas Fir
Hawse pipes
Marine borer hazard
Mooring
Pier pilings
Preservative treatment
Southern Pine
Vessel

6.6 Hawse pipes. Hawse pipes as specified in 3.7 and figure 1 constitute one design which may or may not be acceptable to Commands utilizing this acquisition document. Commands desiring to utilize other hawse pipe designs should specify the design as a deviation to the basic specification.

6.7 Lathe-turned logs. In specifying lathe-turned log camels, it must be noted that the lathe will likely remove a considerable portion of the sapwood. The result is difficulty in reaching desired preservative retention levels, even when the log is incised as described in 3.5.

Preparing Activity:
Navy - YD

(Project 1945-N061)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-28628C	910708

3. DOCUMENT TITLE

CAMEL, WOOD, MARINE, SINGLE LOG CONFIGURATION

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
C. FENAROLI	(805) 982-5604	551-5604
c. ADDRESS (Include Zip Code) CIVIL ENGINEERING SUPPORT OFFICE (1564F) NAVAL CONSTRUCTION BATTALION CENTER PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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